Chapter 5 · Straight line graphs

Starter 2

1. Find each intersection.

1(a)
$y = 4x + 7, \quad 3y = 2x - 1$

1(a)

1(b)
$y = 5x - 1, \quad 3x + 7y = 11$

1(b)

1(c)
$2x - 5y = -1, \quad 5x - 7y = 14$

1(c)

2. Simplify each surd.

2(a)
$\sqrt{80}$

2(a)

2(b)
$\sqrt{200}$

2(b)

2(c)
$\sqrt{125}$

2(c)

3. Make y the subject.

3(a)
$6x + 3y - 15 = 0$

3(a)

3(b)
$2x - 5y - 9 = 0$

3(b)

3(c)
$3x - 7y + 12 = 0$

3(c)

Chapter Overview

Gradients and equations

Use points, gradients and intercepts to describe straight lines.

Parallel and perpendicular lines

Compare gradients and find unknown line equations.

Lengths and areas

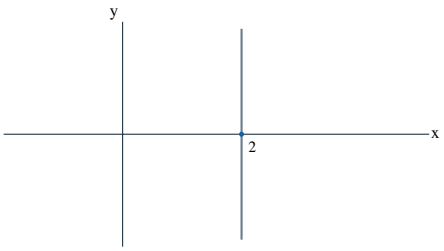
Use intersections, Pythagoras and triangle areas.

Linear modelling

Interpret gradients and intercepts in practical situations.

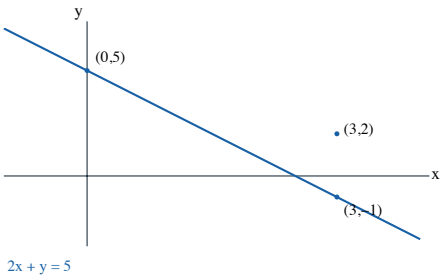
Vertical lines

What equation describes this vertical line?



Equation

Which points lie on $2x + y = 5$?



a

$(0, 5)$

a

b

$(3, 2)$

b

c

$(3, -1)$

c

Examples 6

1 The point (5,a) lies on $y = 3x + 2$. Find a.

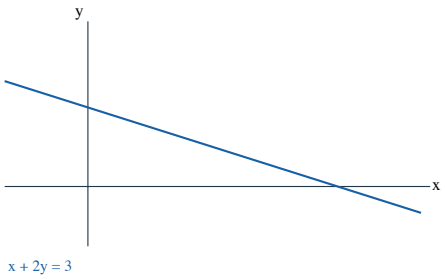
1

2 Find where $2x + y = 5$ meets the x-axis.

2

Test Your Understanding 7

Find both axis intercepts of $x + 2y = 3$.



x + 2y = 3

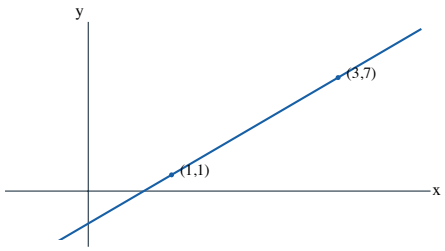
Intercepts

On the x-axis, y=0. On the y-axis, x=0. State an intercept as a coordinate pair.

Recap of gradient 8

Gradient measures the change in y for each unit change in x.

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$



Example The rise is 6 and the run is 2.

Example

Subtract the coordinates in the same direction in numerator and denominator. A vertical line has undefined gradient.

Examples 9

a Find the gradient through $(1,4)$ and $(3,10)$.

a

b Find the gradient through $(5,7)$ and $(8,1)$.

b

c Find the gradient through $(2,2)$ and $(-1,10)$.

c

d Are $(3,4)$, $(5,5)$ and $(11,8)$ collinear?

d

Further Example 10

Find a The line through $(2,-5)$ and $(4,a)$ has gradient -1.

Find a

$y = mx + c$

m is the gradient; c is the y-intercept.

Why c? Set x=0.

Why c.

Why m? Increase x by 1.

Why m.

Example Find the gradient and y-intercept of $4x - 3y + 5 = 0$.

Example

Keep exact fractions rather than rounding.

Standard form Write $y = \frac{1}{3}x - \frac{2}{3}$ in the form ax+by+c=0 with integer coefficients.

Standard form

Just for your interest. . . 14

Why use standard form?

$ax + by + c = 0$

It includes vertical lines such as $x + 4 = 0$, which cannot be written as y=mx+c.

Further mathematics connection

The vector $\binom{a}{b}$ is normal (perpendicular) to the line. For example, $\binom{2}{1}$ is normal to $2x + y = 4$.

$$\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + c = 0$$

In three dimensions, $ax + by + cz + d = 0$ describes a plane with normal vector (a, b, c), provided that vector is nonzero.

Test Your Understanding 15

Your turn Write $y = \frac{2}{5}x + \frac{3}{5}$ in standard form.

Your turn

Exercise 5A – Pure Y1 Page 91 16

9 Show A(2,3), B(4,4), C(10,7) are collinear.

9

10 Show A(−2a,5a), B(0,4a), C(6a,a) are collinear.

10

8 A line meets the axes at (5,0) and (0,2). Find its gradient and standard-form equation.

8

9 Find the gradient and y-intercept of $ax+by+c=0$, where $b\neq0$.

9

10 The line $ax-2y+c=0$ has gradient 3 and y-intercept 5. Find a and c.

10

11 A line through (0,6) has gradient -2. Find its intersection with $5x - 8y - 15 = 0$.

11

12 The lines $y = 3x - 7$ and $ax + 4y - 17 = 0$ meet at $(-3,b)$. Find b and a.

12

Equations using two points/point + gradient 19

GCSE method Find the line through (3,5) with gradient 2.

GCSE method

A new way... 20

A fixed point on a line is (x_1, y_1). A variable point is (x,y).

$m = \dfrac{y - y_1}{x - x_1} \Rightarrow y - y_1 = m(x - x_1)$

Revisit Use the point (3,5) and gradient 2.

Revisit

x_1, y_1, m are constants. Keep x and y as variables. This form is especially useful with fractional gradients.

Quickfire Questions 21

1

$m = 3, \quad P = (1, 2)$

1

2

$m = 5, \quad P = (3, 0)$

2

3

$m = 2, \quad P = (-3, 4)$

3

4

$m = \frac{1}{2}, \quad P = (1, -5)$

4

5

$m = 9, \quad P = (-4, -4)$

5

Use point–gradient form whenever you know a gradient and a point. No expansion is needed unless a particular form is requested.

Using 2 points 22

Example Find the line through (4,5) and (6,2), in standard form.

Example

Your turn Find the line through $(-1,9)$ and (4,5), in standard form.

Your turn

3 Find the line through A(7,2), B(9,−8) in standard form.

3

5 The line through (a,4), (3a,3) is x+6y+c=0. Find a,c.

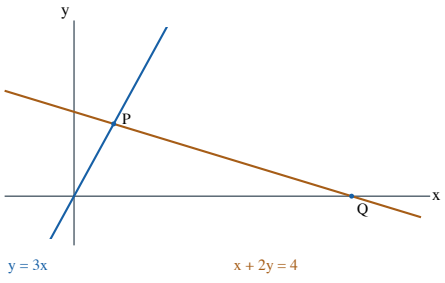
5

6 The line through (7a,5), (3a,3) is x+by−12=0. Find a,b.

6

Intersection of lines

The lines $y = 3x$ and $x + 2y = 4$ meet at P. The second meets the x-axis at Q.



a Find P.

a

b Find Q.

b

Test Your Understanding 25

a Find the equation of L_1 through $(-1,3)$, $(11,12)$. Use integer coefficients.

a

b Find its intersection with $L_2 : 3y + 4x - 30 = 0$.

b

Exercise 5D – Pure Y1 Page 96 26

6 Find where the line through $(-3,-5)$ and $(4,9)$ meets the y-axis.

6

9 The lines y=4x−10 and y=x−1 meet at T. Find the line through T with gradient $-2/3$.

9

14 Find the line through C(0,5) and D(−5,0).

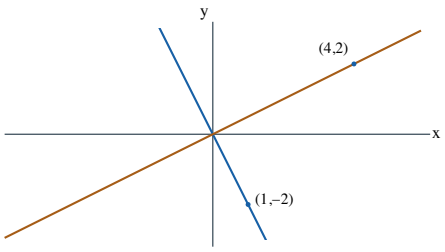
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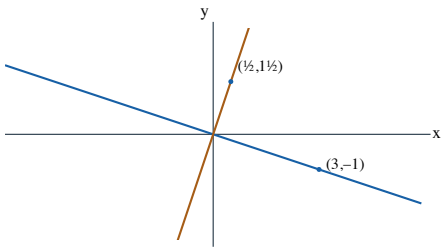
16 The lines y=−2x+1 and y=x+7 meet at L. Find the line through L and M(−3,1).

16

Gradients of perpendicular lines 28

Compare the gradients of these perpendicular pairs.





Pair 1 Calculate both gradients.

Pair 1

Pair 2 Calculate both gradients.

Pair 2

If one rise/run is b/a, a 90° rotation gives −a/b. The gradients are negative reciprocals. This formula requires nonvertical, nonhorizontal lines.

Perpendicular Lines 29

Parallel lines have equal gradients. For perpendicular lines with defined gradients:

$m_1 m_2 = -1$

a

$m = 2$

a

b

$m = -3$

b

c

$m = \frac{1}{4}$

c

d

$m = 5$

d

e

$m = -\frac{2}{7}$

e

f

$m = \frac{7}{5}$

f

A horizontal line is perpendicular to a vertical line; handle this case directly.

Example Problems 30

1 Find the line through (9,10) perpendicular to y=3x+2.

1

2 L_1 passes through A(1,3), B(3,−1). A perpendicular line through B meets the x-axis where?

2

3 Are $y = \frac{1}{2}x$ and 2x−y+4=0 parallel, perpendicular, or neither?

3

Test Your Understanding 31

1 A line through the origin is perpendicular to $y = -\frac{1}{2}x + 4$. Find their intersection A.

1

2 Find the line through (4,7) perpendicular to y=2x+2. Use standard form.

2

Exercise 5E/5F – Pure Y1 Page 96 32

3 A(−6,2), B(4,8), C(6,1), D(−9,−8) form a quadrilateral. Show it is a trapezium.

3

7 Find the line through the origin parallel to 2x−3y−1=0.

7

9 Show the lines through $(-3,0)$, $(3,-2)$ and through $(1,8)$, $(-1,2)$ are perpendicular.

9

10 Show A$(-1,5)$, B$(7,1)$, C$(5,-3)$, D$(-3,1)$ form a rectangle.

10

1. MAT Reflect $(3,4)$ in $3x+4y=50$.

1. MAT

2. MAT Reflect $(1,0)$ in $y=mx$.

2. MAT

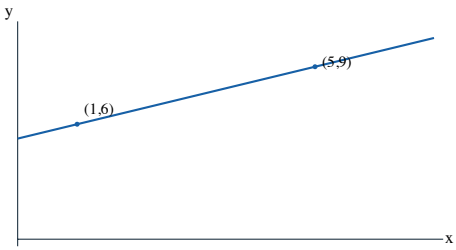
3. STEP For $A=(p_1,q_1), B=(p_2,q_2), C=(p_3,q_3)$, find where the medians from A and B meet and verify this point lies on the median from C. Let $H=(p_1+p_2+p_3, q_1+q_2+q_3)$. Show that AH perpendicular to BC implies $p_2^2+q_2^2=p_3^2+q_3^2$. Give the analogous result for BH perpendicular to AC, then deduce CH perpendicular to AB.

3. STEP

Distances between points 35

Find the distance between (1,6) and (5,9).

Distance



$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Use Pythagoras on the horizontal and vertical changes.

Examples 36

a Find the distance between (3,4) and (5,7).

a

b Find the distance between (5,1) and (6,−3).

b

c Find the distance between (0,−2) and (−1,3).

c

d Find the distance between (1,10) and (4,14).

d

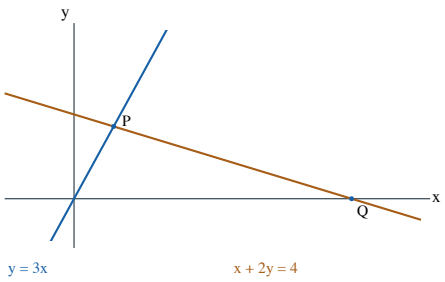
e Find the distance between (3,−1) and (0,1).

e

f Find the distance between (−4,−2) and (−12,4).

f

Lines $y=3x$ and $x+2y=4$ meet at P. Q is the second line's x-intercept; O is the origin.



a Find P.

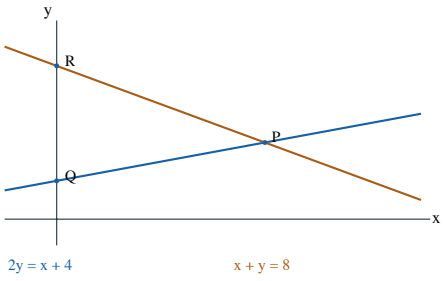
a

b Find the area of triangle OPQ.

b

Further Example 38

Lines $2y=x+4$ and $x+y=8$ meet at P. Their y-intercepts are Q and R respectively.



a Find the length PQ.

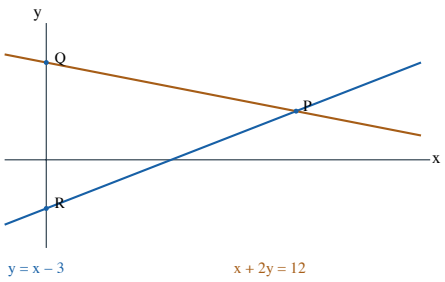
a

b Find the area PQR.

b

A vertical or horizontal side often makes a convenient base for a triangle.

Lines $y=x-3$ and $x+2y=12$ meet at P. $Q=(0,6)$, $R=(0,-3)$ are their y-intercepts.



a Find P.

a

b Find the area PQR.

b

c Find PQ.

c

7 P lies on $y=4-3x$ and is $\sqrt{34}$ units from the origin. Find both possible positions.

7

8 $A(2,7)$, $B(5,-6)$, $C(8,-6)$. Find the side lengths, classify the triangle and find its area.

8

a l_1 passes through $(-4,14)$ with gradient $-1/4$. Find its equation.

a

b $l_2 : y = 3x$ meets l_1 at B. Find B.

b

c l_1 meets the y-axis at A. Find the area OAB.

c

MAT Find the shortest distance from the origin to 3x+4y=25.

MAT

What's the point of straight line equations? 43

Linear relationships

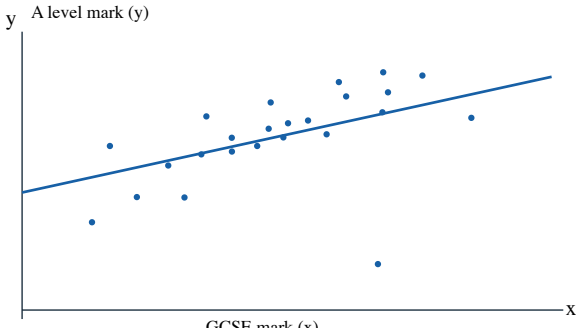
Examples include a fixed salary plus commission, temperature conversions and simple altitude–temperature models.

Arithmetic sequences

The sequence 2,5,8,11,14 has constant difference 3. Its nth term is $3n - 1$, a linear expression.

Look for a constant absolute change. Real data may only be approximately linear; specify the range where a model is reasonable.

Modelling 44



Illustrative line of best fit

A straight-line model

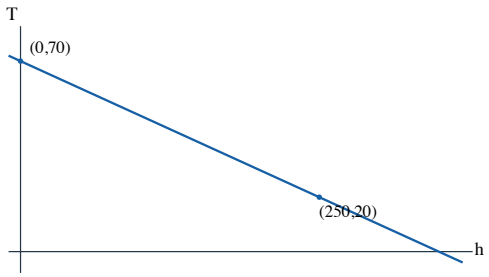
Plot paired GCSE and A level marks on a scatter diagram. A line of best fit can describe an approximate relationship.

Interpret the parameters

The gradient estimates the change in predicted A level mark per extra GCSE mark. The intercept is the prediction when the GCSE mark is zero.

The intercept may have no practical meaning if zero lies far outside the data. Predictions are estimates; extrapolation beyond the observed range is less reliable.

Example 45



Temperature against altitude

a Temperature is 70˚F at altitude 0 m and 20˚F at altitude 250 m. Find a linear model T=mh+c.

a

b Interpret m and c.

b

c At what altitude does the model predict 0˚F?

c

This prediction extrapolates beyond the two given observations.

Evaluating if a linear model is most suitable 46

Population A population is 26,000 in 2017 and increases by 150 each year. Model it t years after 2017.

Population

Assumption When is a linear model reasonable?

Assumption

Electricity cost E (£) for usage h (kWh):

h	0	15	40	60	80	110
E	45	46.8	49.8	52.2	54.6	58.2

a. Draw a graph of the data.

Sketch

b Why is a linear model appropriate?

b

c Find an equation E=ah+b.

c

d Interpret the coefficients.

d

e Find the cost for 65 kWh.

e

a 20˚C corresponds to 68˚F; 9˚C to 48.2˚F. Find F=mC+c.

a

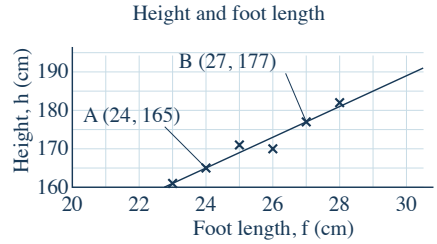
b Convert a recorded temperature of 101.3˚F to Celsius.

b

c At which temperature do the scales agree?

c

8. The scatter graph shows height h (cm) and foot length f (cm) for a group of students.



8(a) Why is a linear model suitable?

8(a)

8(b) Use A(24,165), B(27,177) to find h=af+b.

8(b)

8(c) Predict h when f=26.5.

8(c)

9. Demand: $P = -\frac{3}{4}Q + 35$. Supply: $P = \frac{2}{3}Q + 1$.

a. Sketch both lines on the same axes.

Sketch

9(b) Find the equilibrium price P and quantity Q.

9(b)